25900651 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCSTA203 - STATISTICAL RESEARCH TECHNIQUES USING SOFTWARES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. What you meant by pilot survey? [U] / [CO1]
2. Mention two applications of research in Education. [K] / [CO1]
3. What is qualitative research design? [U] / [CO1]
4. Define type 1 error and significance level. [U] / [CO1]
5. Define statistic. [U] / [CO1]
6. Mention any two assumptions in Z-test. [U] / [CO2]
7. Mention one advantage and one disadvantage of non parametric test. [U] / [CO4]
8. What is the assumption about the probability distribution of population in t test? [K] / [CO3]
9. Mention two important applications of t distribution in small sample test. [A] / [CO3]

10. State the null and alternate hypothesis in the analysis of variance. [U] / [CO3]

[2x7 = 14]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **6** marks

11. Define cluster sampling. Discuss the advantages and disadvantages of it. [U] / [CO1]

12. Explain exploratory research design with an example . [U] / [CO1]

13. Explain the term critical region . Differentiate between the critical regions of a one tailed test and a two tailed test . [U] / [CO1]

14. Write the procedure for testing a hypothesis. [U] / [CO1]

15. An IQ test was administered to 5 persons before and after they were trained. The results are as follows: [An] / [CO3]

Candidate	A	B	C	D	E
I.Q before training	110	120	123	132	125
I.Q. after training	120	118	125	136	121

Test whether the training was effective?

16. The following data is about marital status and performance in exam. Examine whether performance in exam is independent of marital status. [A] / [CO4]

	good performance	bad performance
married	60	80
unmarried	20	40

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. What are the precautions for writing research reports ? [U] / [CO1]

18. A researcher wants to test whether a new teaching method improves student performance. 27 students take a math test before and after the new method is implemented. The researcher records their scores [A] / [CO4]

before and after the intervention. Use Sign test at 5% level of significance to check whether there is a statistically significant difference in student performance after using the new teaching method?

Students Test Scores(Before and After)

Before: 70, 68, 85, 90,76, 82, 65, 76, 74, 88, 86,78,76, 89, 90, 85, 84, 83, 70, 73, 77, 87,67, 90, 85, 72, 72

After : 75, 72, 84, 91, 80, 83, 70, 78, 77, 75, 78, 89, 69, 80, 91, 87, 83, 85, 79, 76, 79, 85, 64,91,90,69, 68

[12x1 = 12]